

Name: _____



Expressing an integer as a product of its prime factors

Task A: Composite and prime numbers

1) Draw arrays to prove that the following are **composite** numbers.

- a) 18 b) 32 c) 25 d) 51

Task B: Product of prime factors

1) Write the following as a product of their **prime factors**.

- a) 78 b) 165

- c) 525 d) 840
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Task C: Product of prime factors using index notation

1) Fill in the missing digits.

a) $2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 7 = 2^{\square} \times 3^{\square} \times 7^{\square}$

b) $2 \times 2 \times 5 \times 5 \times 5 \times 11 \times 11 = 2^{\square} \times 5^{\square} \times 11^{\square}$

c) $5 \times 13 \times 5 \times 5 \times 3 \times 13 \times 3 \times 5 = 3^{\square} \times 5^{\square} \times 13^{\square}$

d) $17 \times 2 \times 17 \times 7 \times 7 \times 2 \times 17 \times 7 \times 17 = 2^{\square} \times 7^{\square} \times 17^{\square}$

e) $5 \times 2 \times 5 \times 7 \times 7 \times 11 \times 5 \times 7 \times 11 = 2 \times 5^{\square} \times 7^{\square} \times 11^{\square}$

f) Why is there no box for the 2 in e?

2) Use $60 = 2^2 \times 3 \times 5$ to write the following as a product of their **prime factors**.

- a) 120
 - b) 180
 - c) 300
 - d) 600
 - e) 6000
 - f) 4200
 - g) 30
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